

1. Participant Information & Topic Name

- Team Name [to fill]
- Team Lead Name [to fill]
- Team Lead Contact No [to fill]

Selected Track

- ☐ Quantum Machine Learning (QML)
- **Post-Quantum Cryptography (PQC)**
- ☐ Web3 & Blockchain
- ☐ Generative AI & Machine Learning

Q-SHIELD

Continuous device trust with post-quantum-signed evidence

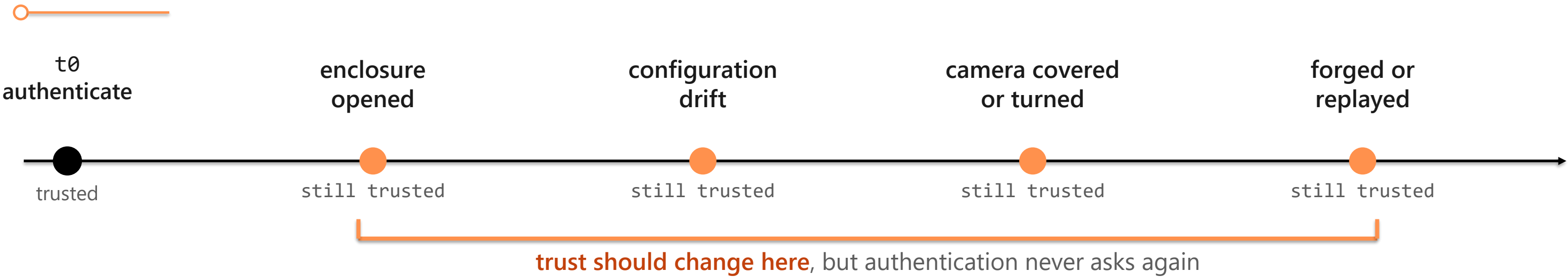
Authentication answers **who are you** once.
Q-SHIELD keeps asking **can you still be trusted.**



ML-KEM-768 and ML-DSA-65 (FIPS 203, 204)

2. Problem Definition

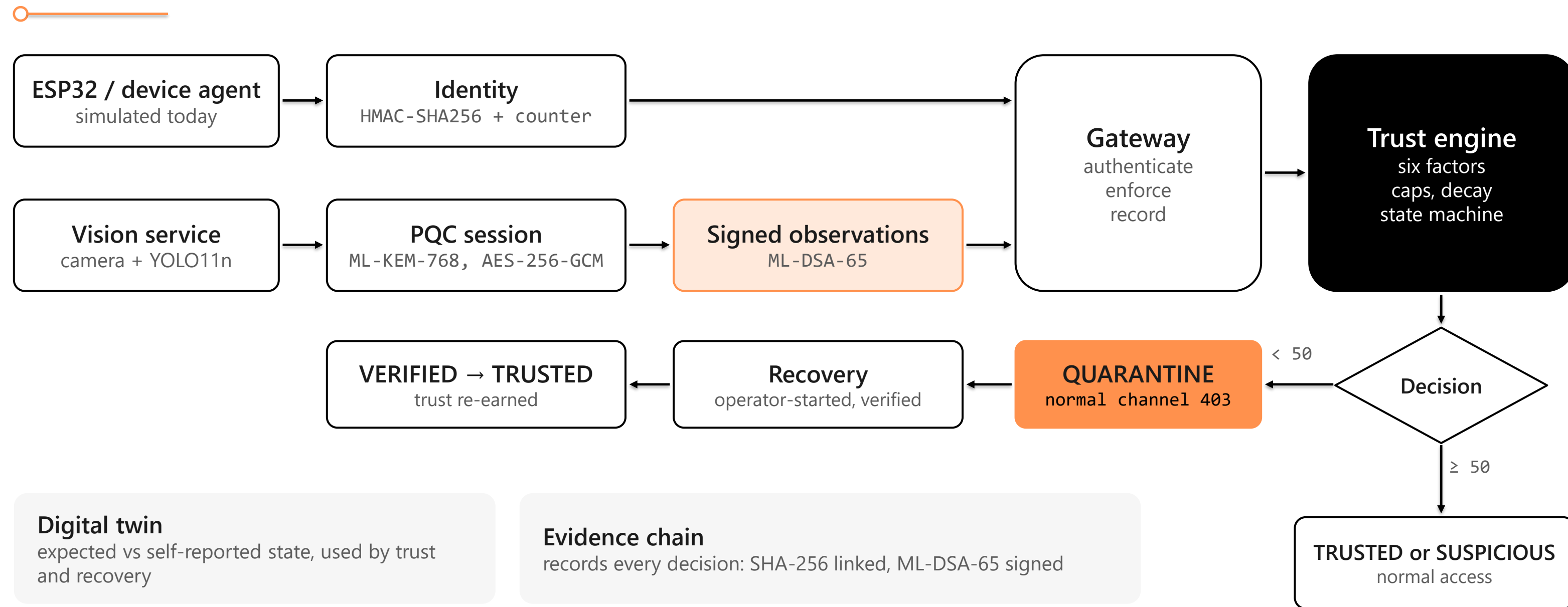
Problem description: authenticated is not the same as trustworthy



Threat	Authentication alone	What Q-SHIELD adds
Forged or replayed message	rejected, then forgotten	bounded pressure (≤ 25); 3 replays in 10 min hold the score at 65
Enclosure opened	nothing to see	physical factor CRITICAL; score held at 55
Camera covered, frozen or turned	nothing to see	signed camera-health report; visual factor HIGH
Tamper + visual evidence within 60 s	nothing to see	confirmed incident: score 30, quarantine
Quantum adversary recording traffic	classical signatures forgeable later	ML-DSA-65 signs the evidence; ML-KEM-768 keys the session

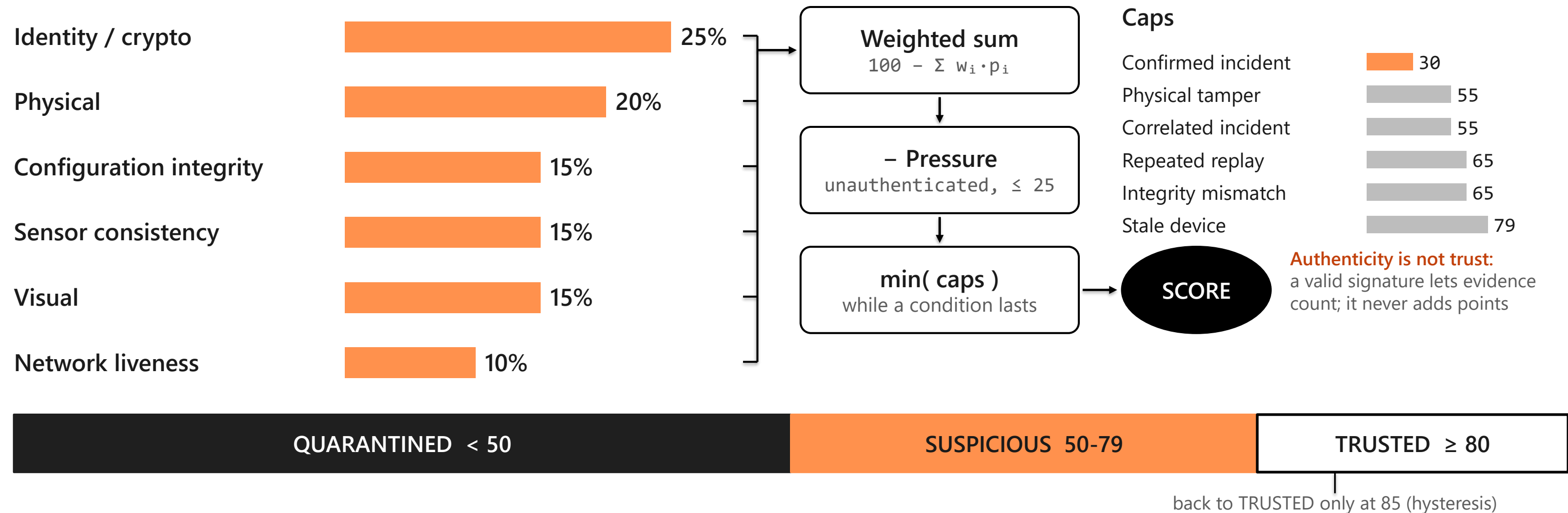
3. Proposed Solution & Execution Plan

Working principle: observe → verify → score → enforce → recover → prove, for every device



3. Proposed Solution & Execution Plan

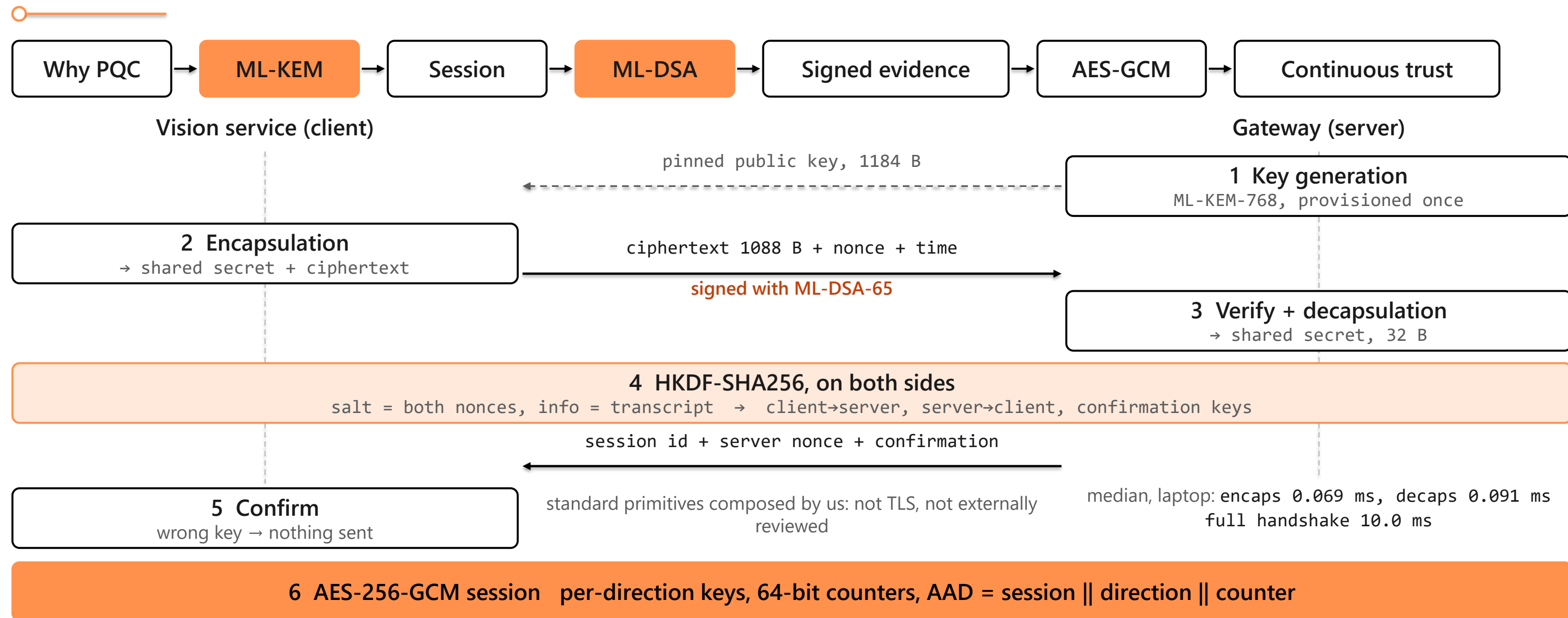
Core functionality: continuous six-factor trust



Recovery credit $\Delta c = \text{clamp}(t - \max(t_{\text{prev}}, t_{\text{violation}}), 0, 45 \text{ s})$
only clean authenticated evidence earns time; silence never heals a device

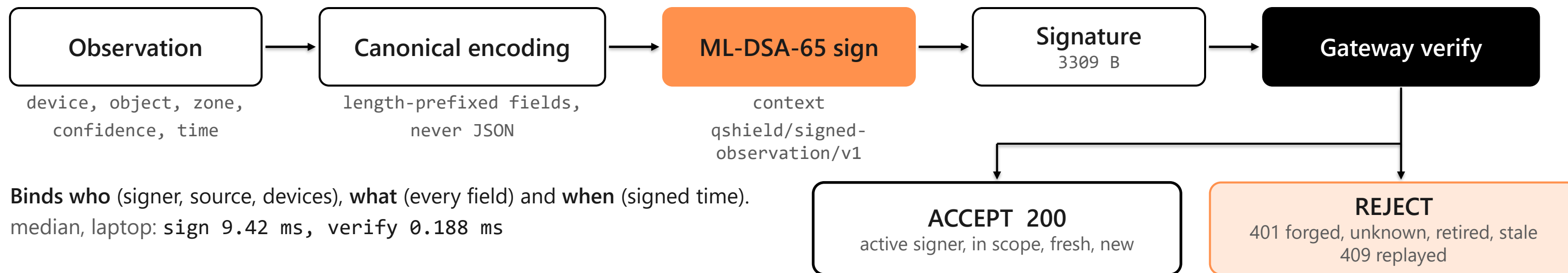
4. Technical Design

Post-quantum session: ML-KEM-768 key exchange, ML-DSA-65 signed, AES-256-GCM data

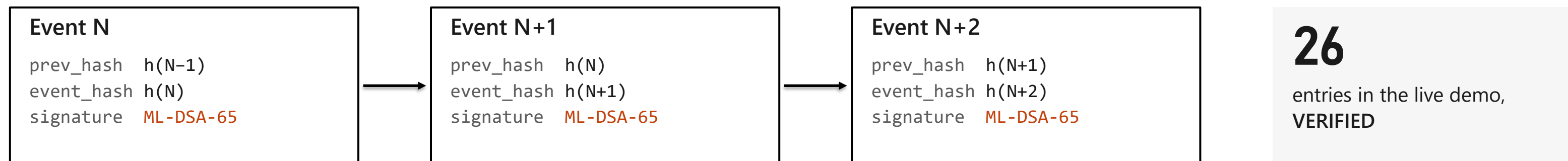


4. Technical Design

ML-DSA-65 signed observations and evidence chain



Evidence chain: every decision, SHA-256 linked and ML-DSA-65 signed

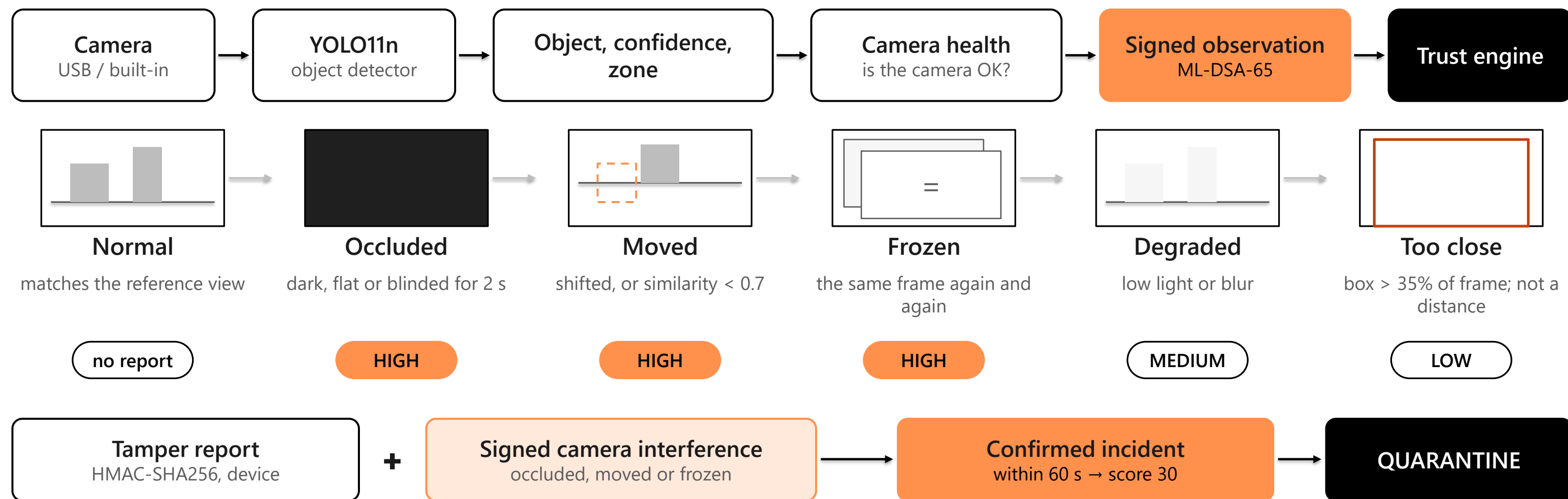


Verification recomputes every hash and checks every signature: an edited, reordered or removed entry is found.

Not anchored externally: cutting the tail needs an outside copy of the head; holding the database and the key together would allow a rewrite.

4. Technical Design

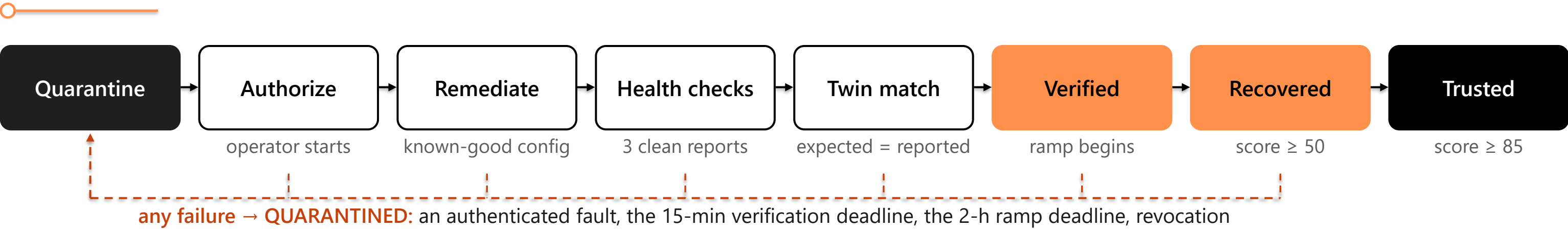
AI vision security: the camera is part of the boundary



Camera alone: a penalty, never quarantine. Camera + rule match from the same signer: one modality, so a stolen vision key cannot confirm an incident. Rule match = restricted class in a restricted zone (a policy, not a verdict). Live on a USB webcam, CPU, about 5 frames/s; detection accuracy not measured.

4. Technical Design

Quarantine, digital twin and self-healing recovery



Digital twin: expected (operator) vs reported (authenticated)

Field	Expected	Reported	Verdict
Firmware	agent-0.1	agent-0.1	MATCH
Configuration	cfg-good-1	cfg-tampered	MISMATCH
Capabilities	tamper, temp, vib.	same three	MATCH
Temperature	-20 to 60 °C	27.4 °C	MATCH
Tamper state	closed	open	MISMATCH

Quarantine at the gateway

	Normal channel	Recovery channel
TRUSTED, SUSPICIOUS, RECOVERED	allowed	not open
QUARANTINED, RECOVERING, VERIFIED	403 blocked	allowed
Forged message	401	401

Enforced after authentication: forged traffic can neither trigger quarantine nor bypass it.

4. Technical Design

Technology stack and implementation plan

Each technology with the role it plays

Cryptography

pqcrypto ML-KEM-768, ML-DSA-65
cryptography AES-256-GCM, HKDF
hashlib, hmac SHA-256 chain, HMAC

AI / vision

OpenCV camera capture
YOLO11n object detection
NumPy camera-health checks

Backend

FastAPI, uvicorn gateway API
Pydantic strict schemas
Trust engine pure Python

Frontend

ES modules dashboard, no build
Presentation one-screen live view

Database

SQLite events, observations, twin, evidence chain, trust state

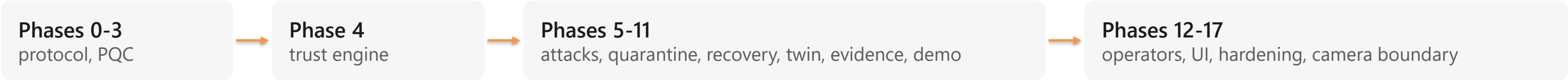
Deployment

Laptop gateway optional TLS, named operators, roles

Hardware (planned)

ESP32 firmware skeleton, not compiled
BME280, MPU6050, reed switch sensors, tamper

Implementation plan: built and tested in 17 phases



5. Feasibility & Innovation

Technical feasibility and performance: what is real, what is not yet, what was measured



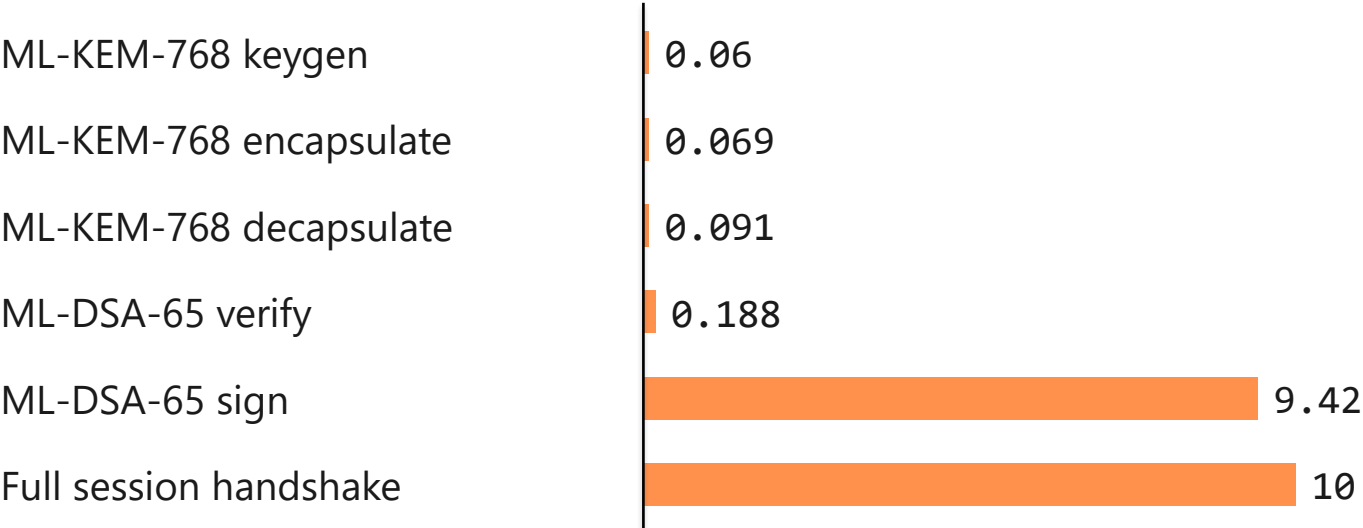
Real and tested

- ML-KEM-768, ML-DSA-65, AES-256-GCM, HKDF
- SHA-256 linked, signed evidence chain
- YOLO11n vision on a real USB webcam
- trust, quarantine and recovery engines
- digital twin, dashboard, camera preview

Simulated or not yet

- device telemetry: labelled software agent
- ESP32 never compiled; no physical sensors
- no attestation; camera thresholds unvalidated
- single gateway (SQLite); chain not anchored

Post-quantum operations, median ms (laptop CPU)



Key exchange is about 100x cheaper than signing; nothing measured on an ESP32.

785

Python tests
passing

40

JavaScript tests
passing

70

NIST ACVP
vectors agree

53 μs

per trust-engine
update (mean)

5. Feasibility & Innovation

Innovation / USP

What one-time authentication cannot do, and how it holds up under abuse

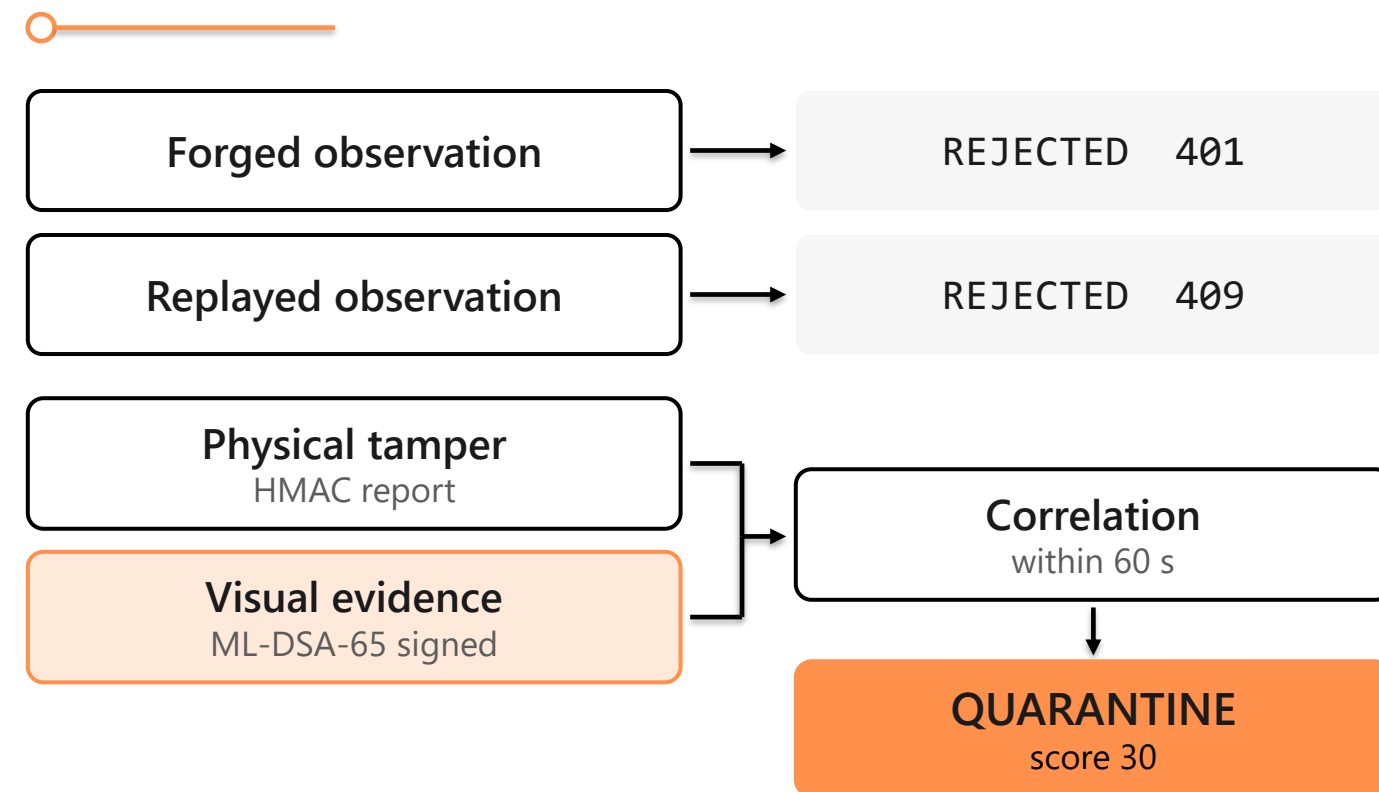
- Continuous, explainable trust**
every change lists the reasons that sum to it
- Authenticity is not trust**
signatures decide what counts; they never add points
- Camera as a security boundary**
covered, frozen or turned is signed evidence
- Bounded grieving**
forged traffic is capped at 25 points: never quarantine
- Earned recovery, signed evidence**
post-quantum signatures on every decision

Attack (tested)	Result
Replayed handshake	one-use nonce: one session only
Replayed or reordered message	strict counters: rejected
Rejection flood	300 forged → 8 rows; trust unchanged
Persistent refusals	30 refusals → 1 session (was 30)
Session exhaustion	cap 64; 20 handshakes/h; backoff
Forged 'all clear' in quarantine	401; stays QUARANTINED

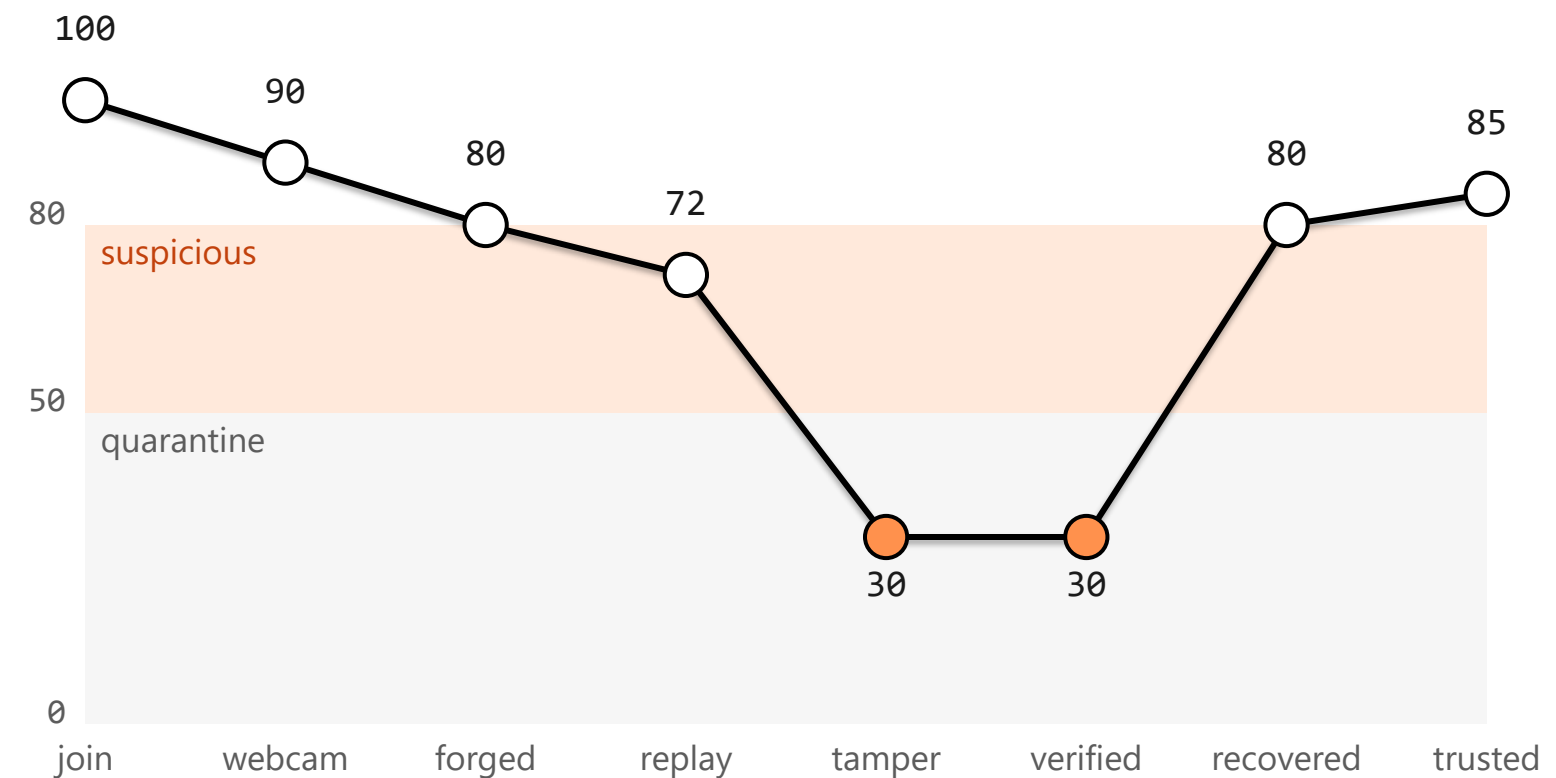
The novelty is the loop, not one algorithm. A prior-art survey has not been done: we claim an integration of standard mechanisms.

6. Expected Outcome

Expected results: the live demo, end to end, and what comes next



Attacks are simulated and labelled; the webcam detections are real.



Live run on the USB webcam, 2026-10-02; recovery in announced time-lapse

Future scope, with hardware

- compile and flash the ESP32; match the 7 test vectors
- reed switch, then BME280 and MPU6050
- stage camera tests, calibrate; anchor the evidence head

Real-world impact

- labs, server rooms, critical sites: devices must keep earning trust
- every incident arrives with a signed, explainable trail

Authenticate once. Verify continuously. Recover only with evidence.